Fiscal Year:	FY 2024	**Task Last Updated:**	FY 01/30/2024
PI Name:	Bouxsein, Mary Ph.D.		
Project Title:	The Effects of Spaceflight and Reloading on Skeletal Muscle and Bone		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	None		
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Zip Code:	02215-5400	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2018 Space Biology NNH18ZTT002N:Russian Bion-M2 Mission
Start Date:	10/01/2020	**End Date:**	09/30/2023
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:	Bion-M2		
Flight Assignment:			
Key Personnel Changes/Previous PI:	No changes		
COI Name (Institution):	Rutkove, Seward M.D. (Beth Israel Deaconess Medical Center, Inc./Harvard Medical School)		
Grant/Contract No.:	80NSSC20K1147		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Due to the short-term nature of prior rodent studies (i.e., 2 weeks of spaceflight), the musculoskeletal response to longer duration spaceflight remains ill-defined. Moreover, though highly informative, prior animal studies examining the effects of reloading on skeletal muscle and bone have only assessed one or the other tissue separately, rather than the interrogation of possible muscle-bone crosstalk and associated molecular mechanisms. Thus, our overall objectives are to assess musculoskeletal response and readaptation to the Earth's gravity following long-term spaceflight and to advance knowledge regarding the underlying physiological and molecular mechanisms contributing to bone and muscle recovery and their interactions. We aim to assess functional, structural, and molecular alterations in skeletal muscle and bone during reloading and identify candidate mechanisms that will lead to better recovery strategies following long-term spaceflight and disuse-induced osteoporosis on Earth. We propose to analyze skeletal muscle and bone microarchitecture and function using histology, muscle dynamic testing, electrical impedance myography, microcomputed tomography, static histomorphometry, reference point indentation, and fracture toughness. In addition by using a candidate gene approach along with unbiased whole transcriptomic analyses, we will be able to test our hypotheses that inflammatory and autophagic responses precede the canonical Wnt and mTOR signaling pathways to promote bone and skeletal muscle regeneration during reloading and identify novel unexplored mechanisms that may lead to new therapeutic interventions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Reduced mechanical loading due to injury and/or disuse leads to deterioration of the musculoskeletal system. On Earth, this reduced mechanical loading leads to bone and muscle loss, and increased risk of musculoskeletal injury. By studying the musculoskeletal changes in mice exposed to long-duration spaceflight and then allowed to re-adapt to Earth's gravitational loading, we aim to gain insights into both how to present musculoskeletal deterioration during disuse and how to enhance the recovery of these tissues after a period of disuse.
Task Progress:	This study was never initiated, per NASA. [NOTE: Per the NASA Space Biology Program, these investigations were discontinued (Ed., 5/14/25).]
Bibliography Type:	Description: (Last Updated: 04/02/2026)